SDG BLOCKCHAIN ACCELERATOR

Impact Measurement Framework

1. Project Information

Project Name: Results-Based Payments for Meteorological Data (RBP-MetData)
Solution Maker: ClimaFi
Challenge Owner: UNDP Office of Procurement
Geographic Scope: Pilot (e.g., Tanzania – NMHS / TMA)

2. Theory of Change

Problem

Meteorological infrastructure suffers from:

- underfunded maintenance
- inconsistent data quality
- lack of incentive alignment
- absence of verifiable performance tracking

Intervention

ClimaFi introduces:

- automated verification of meteorological data
- tokenized proof of compliance
- a shadow results-based payment system
- transparent dashboards for accountability

Outcome Logic

Stage	Description
Input	Meteorological data streams (stations)
Activity	Verification + tokenization + shadow valuation
Output	Verified datasets + receipt tokens
Outcome	Improved data quality + visibility + incentives
Impact	Sustainable financing of climate infrastructure

3. SDG Alignment

SDG	Target	Indicator	Contribution
SDG 13 – Climate Action	13.1	13.1.2	Strengthens early warning systems via reliable data
SDG 9 – Infrastructure	9.1	9.1.1 (adapted)	Improves resilience of data infrastructure
SDG 17 – Partnerships	17.6	17.6.1	Enables data-driven climate finance collaboration

4. Impact Measurement Architecture

The framework measures impact across **4 layers**:

4.1 Layers Overview

Layer	What is measured
Input	Data ingestion & system usage
Output	Verified datasets & tokens
Outcome	Behavioral + system improvements
Impact	Structural change in financing + resilience

5. Key Performance Indicators (KPIs)

5.1 Input KPIs (System Usage)

KPI	Definition	Measurement Method	Target

Station Connectivity	Stations actively sending data	Count of active stations	≥ 90% of pilot
Data Submission Volume	Number of datasets received	API ingestion logs	Increasing trend
System Uptime	Availability of backend services	Monitoring tools	≥ 99%

5.2 Output KPIs (Verification & Proof)

KPI	Definition	Measurement Method	Target
Verified Dataset Rate	% of submissions passing verification	Approved / Total submissions	≥ 75%
Token Issuance Volume	Number of Data Receipt Tokens minted	Blockchain records	Increasing trend
Verification Latency	Time from submission → verification	Timestamp difference	< 60 min
Data Integrity Score	Hash match consistency	On-chain vs off-chain validation	100%

5.3 Outcome KPIs (Behavioral & System Change)

KPI	Definition	Measurement Method	Target
Data Quality Improvement	Increase in verified dataset rate over time	Month-over-month comparison	+20% improvement
Station Reliability	Reduction in data gaps	Missing vs expected submissions	↓ 30% gaps
Operator Responsiveness	Time to fix rejected submissions	Submission retry time	↓ over time

Transparency Index	% of datasets traceable via tokens	Tokenized / verified datasets	100%

5.4 Impact KPIs

KPI	Definition	Measurement Method	Target
Financial Readiness	Accuracy of shadow payouts vs expected	Model validation	≥ 95% accuracy
Cost of Verification	Reduction in manual audit cost	Baseline vs PoC	↓ significant
Funding Efficiency	Funds allocated per verified dataset	Shadow model	Improved ratio
Institutional Adoption	Stakeholder usage of dashboards	Active users	≥ defined stakeholders

6. Core Metrics Definitions (Technical)

6.1 Verification Rate

Verification Rate = (Approved Submissions / Total Submissions) × 100

6.2 Data Quality Score

Data Quality Score =
(Timeliness Weight × Timeliness Score) +
(Completeness Weight × Completeness Score) +
(Accuracy Weight × Range Validity Score)

Example weights:

- Timeliness: 40%
- Completeness: 30%
- Accuracy: 30%

6.3 Verification Latency

Latency = Verification Timestamp - Submission Timestamp

6.4 Shadow Payment Value

Shadow Value = Base Rate × Data Quality Score × Frequency Multiplier

6.5 Station Reliability

Reliability = (Actual Submissions / Expected Submissions) × 100

7. Measurement Methodology

7.1 Data Sources

Source	Description
API Logs	Submission timestamps and payloads
Verification Engine	Pass/fail decisions + reasons
Blockchain	Token issuance + metadata
Dashboards	Aggregated metrics
Database	Historical performance

7.2 Data Flow

Submission → Verification → Hash → Token → Dashboard → Metrics Engine

All metrics derive from:

- backend logs
- verification outputs

- blockchain proofs

8. Monitoring & Reporting Plan

8.1 Reporting Frequency

Frequency	Report Type
Real-time	Dashboard metrics
Weekly	Technical performance
Monthly	Stakeholder reports
End of Pilot	Impact evaluation report

8.2 Reporting Outputs

1. Live Dashboard

- verification rate
- token issuance
- station performance
- shadow earnings

2. Monthly Report

- KPI tracking
- performance trends
- identified issues
- recommendations

3. Final Impact Report

- baseline vs post-pilot comparison
- readiness for live payments
- cost-benefit analysis

9. Baseline vs Target Measurement

Metric	Baseline	Target
Verified Data Rate	Low / inconsistent	≥ 75%
Data Gaps	High	↓ 30%
Audit Transparency	Low	100% traceable
Payment Logic	Non-existent	Fully simulated
Trust Level	Low	High (dashboard + proof)

10. Risk & Measurement Limitations

Risk	Impact	Mitigation
Data inconsistency	Skewed KPIs	Validation rules
Connectivity issues	Missing data	Asynchronous validation
Limited pilot scope	Sample bias	Scalable design
No real payments	Financial metrics simulated	Shadow model calibration

11. Learning & Adaptation Framework

Feedback Loops

- Weekly stakeholder calls
- Dashboard-driven insights
- Iterative KPI tuning
- Verification rule adjustments

Adaptive Improvements

- refine quality thresholds

- adjust payout logic
- optimize verification engine
- improve UX for operators

12. Long-Term Impact Measurement (Post-Pilot)

Once live payments are enabled:

Additional Metrics

- actual funds disbursed
- cost per verified dataset
- ROI for donors
- uptime improvement of stations
- insurance / commercial use of data

13. Conclusion

This Impact Measurement Framework enables:

- **Quantifiable validation of data quality**
- **Transparent performance tracking**
- **Simulation of results-based financing**
- **Evidence generation for scaling**